HAT-P-32b

R-1040

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_150925` · created 2026-07-30T07:56:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457290.705 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.079 +/- 0.033
Transit depth (Rp/Rs)^2	0.62 +/- 0.52 [%]
Orbital Inclination (inc)	84.32 +/- 1.77
Airmass coefficient 1 (a1)	1.19 +/- 0.018
Airmass coefficient 2 (a2)	-0.1299 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	None
Optimal Aperture	4.36
Optimal Annulus	6.47
Transit Duration (day)	0.1099 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.079 ± 0.033 vs 0.1489 (deviation 1.56σ)
Duration fit vs published	0.1099 d vs 0.1304 d (deviation 1.57σ)
Mid-time O–C	-4.3 min
Depth SNR	1.8
Residual scatter	1.46 %

Light curve

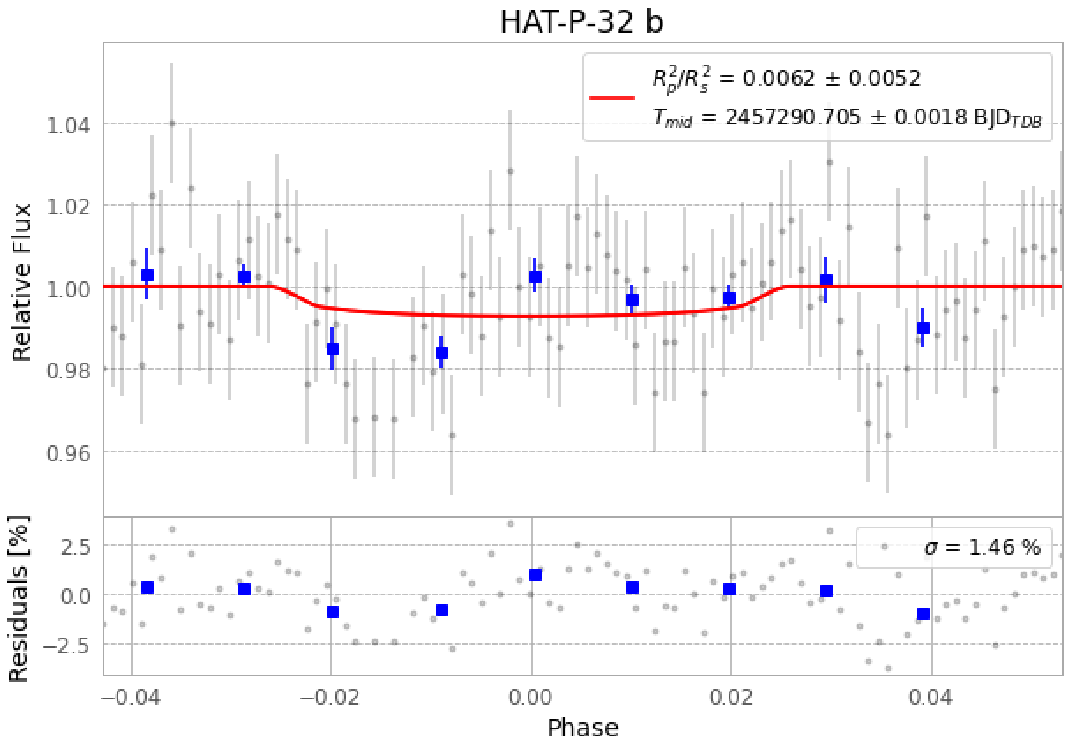

Figure 1 — FinalLightCurve_HAT-P-32 b_2015-09-24.png (SHA-256 3bae5cce29856ab5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [461, 86], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 12c4cf139a47a213…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd`

Data provenance

102 FITS frames (67 MB), `HATP-32150925023012.FITS` → `HATP-32150925073312.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-150925.log (e6a98d148db3…), FinalLightCurve_HAT-P-32 b_2015-09-24.png (3bae5cce2985…), FinalLightCurve_HAT-P-32 b_2015-09-24.csv (898589442c2e…)

Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)